Exam Hall Ticket

[No of Questions: 03] [2612] – 2643PC12 Seat No S26040073

Society for Computer Technology and Research's

PUNE INSTITUTE OF COMPUTER TECHNOLOGY

INSEM EXAMINATION AUG-OCT 2026

S.Y. B.Tech (AI&DS) (SEM: III)

(2643PC12) (Data Structures)

Duration: One Hour **Maximum Marks: 20**

Instruction to the Candidates.

1. *Attempt Any Two Questions of Three Questions*
2. *Neat diagrams must be drawn wherever necessary.*
3. *[Marks], [Course Outcome] and [Bloom's Taxonomy Level] given at the end of each question.*
4. *Use of non-programmable calculators is allowed.*
5. *Assume suitable data if necessary.*

CO-1 : Demonstrate use of Array to store and process data and understand the computational efficiency of the principal algorithms for searching and sorting

Q.1. A) Two-dimensional arrays can be stored in row-major or column-major order. Compare their memory layouts, address calculation formulas, and applications. [05 Marks] [CO1] [L4]

B) Given [38, 12, 45, 7, 23, 16, 31, 9], illustrate Merge Sort with intermediate subarrays and state its best and worst time complexity. [05 Marks] [CO1] [L3]

Q.2. A) For the array [5, 9, 14, 18, 23, 29, 34, 41, 47, 53, 61, 72, 84], show the step by step how Fibonacci Search is used to find number 47, indicating the Fibonacci numbers and positions examined. [05 Marks] [CO1] [L3]

B) An array organizes elements through indices. Explain the array as an Abstract Data Type (ADT), covering its operations, characteristics, advantages, and limitations with examples. [05 Marks] [CO1] [L2]

Q.3. A) Different data structures serve different computational requirements. Classify them with examples and compare linear and non-linear structures with suitable applications. [05 Marks] [CO1] [L4]

B) A bank maintains the transaction amounts of its customers for a particular day. The transaction amounts are stored in the following order:
4200, 1700, 8900, 3500, 6400, 2300, 7100, 5000, 2800, 9300
The bank wants to arrange these transaction amounts in ascending order using Shell Sort for efficient processing and reporting. Apply Shell Sort to the given list using the gap sequence n/2, n/4, ..., 1. Show the array after each gap-based pass and write the final sorted array. Also state the best-case and worst-case time complexity of Shell Sort. [05 Marks] [CO1] [L3]

********** BEST OF LUCK **********